

Type Conversion (Type Casting)

Type conversion in Python refers to converting the value of one data type to another using **built-in functions** such as `int()`, `float()`, `str()`, `bool()`, `list()`, `tuple()`, `set()`, and `dict()`.

Each section below explores conversions **from a specific data type** to others, with examples and error cases.

1. Converting from `int`

Given:

```
Python
int_a = 2
```

Target Type	Code	Output	Notes
float	<code>float(int_a)</code>	2.0	Valid
str	<code>str(int_a)</code>	'2'	Valid
bool	<code>bool(int_a)</code>	True	Non-zero ints are True
list	<code>list(int_a)</code>	Error	int is not iterable
set	<code>set(int_a)</code>	Error	int is not iterable
tuple	<code>tuple(int_a)</code>	Error	int is not iterable
dict	<code>dict(int_a)</code>	Error	int is not iterable

2. Converting from float

Given:

```
Python
float_n = 3.1
```

Target Type	Code	Output	Notes
int	<code>int(float_n)</code>	3	Truncates decimal part
str	<code>str(float_n)</code>	'3.1'	Valid
bool	<code>bool(float_n)</code>	True	Non-zero float is True
list	<code>list(float_n)</code>	Error	float is not iterable
set	<code>set(float_n)</code>	Error	float is not iterable
tuple	<code>tuple(float_n)</code>	Error	float is not iterable
dict	<code>dict(float_n)</code>	Error	float is not iterable

3. Converting from str

Given:

```
Python
string_c = 'power'
```

Target Type	Code	Output	Notes
int	<code>int('10')</code>	10	Only numeric strings
	<code>int('10.9')</code>	Error	Use <code>float()</code> instead
	<code>int('power')</code>	Error	Invalid literal
float	<code>float('10.8')</code>	10.8	Valid float string

	<code>float('power')</code>	Error	Invalid string
bool	<code>bool(string_c)</code>	True	Non-empty string
list	<code>list(string_c)</code>	<code>['p', 'o', 'w', 'e', 'r']</code>	Converts to list of characters
tuple	<code>tuple(string_c)</code>	<code>('p', 'o', 'w', 'e', 'r')</code>	Valid
set	<code>set(string_c)</code>	<code>{'w', 'o', 'e', 'p', 'r'}</code>	Removes duplicates
dict	<code>dict(string_c)</code>	Error	Needs key-value pair structure

4. Converting from list

Given:

```
Python
list_d = [1, 2, 3, 4]
```

Target Type	Code	Output	Notes
int	<code>int(list_d)</code>	Error	Not allowed
float	<code>float(list_d)</code>	Error	Not allowed
str	<code>str(list_d)</code>	<code>"[1, 2, 3, 4]"</code>	Converts entire list to string
bool	<code>bool(list_d)</code>	True	Non-empty list
tuple	<code>tuple(list_d)</code>	<code>(1, 2, 3, 4)</code>	Valid
set	<code>set(list_d)</code>	<code>{1, 2, 3, 4}</code>	Removes duplicates if any
dict	<code>dict(list_d)</code>	Error	Needs list of key-value pairs

5. Converting from tuple

Given:

```
Python
tuple_f = (1, 2, 3, 4)
```

Target Type	Code	Output	Notes
int	<code>int(tuple_f)</code>	Error	Not allowed
float	<code>float(tuple_f)</code>	Error	Not allowed
str	<code>str(tuple_f)</code>	<code>"(1, 2, 3, 4)"</code>	Converts whole tuple to string
bool	<code>bool(tuple_f)</code>	True	Non-empty tuple
list	<code>list(tuple_f)</code>	<code>[1, 2, 3, 4]</code>	Valid
set	<code>set(tuple_f)</code>	<code>{1, 2, 3, 4}</code>	Valid
dict	<code>dict(tuple_f)</code>	Error	Must be tuple of key-value pairs

6. Converting from set

Given:

```
Python
set_e = {3, 4, 5, 6}
```

Target Type	Code	Output	Notes
int	<code>int(set_e)</code>	Error	Not allowed
float	<code>float(set_e)</code>	Error	Not allowed
str	<code>str(set_e)</code>	<code>"{3, 4, 5, 6}"</code>	Converts whole set
bool	<code>bool(set_e)</code>	True	Non-empty set
list	<code>list(set_e)</code>	<code>[3, 4, 5, 6]</code>	Order may vary

tuple	tuple(set_e)	(3, 4, 5, 6)	Order may vary
dict	dict(set_e)	Error	Must be iterable of key-value pairs

7. Converting from dict

Given:

```
Python
dict_g = {1: 1, 2: 4, 3: 9}
```

Target Type	Code	Output	Notes
int	int(dict_g)	Error	Not allowed
float	float(dict_g)	Error	Not allowed
str	str(dict_g)	"{1: 1, 2: 4, 3: 9}"	Valid
bool	bool(dict_g)	True	Non-empty dictionary
list	list(dict_g)	[1, 2, 3]	Keys only
tuple	tuple(dict_g)	(1, 2, 3)	Keys only
set	set(dict_g)	{1, 2, 3}	Keys only

8. Converting from bool

Given:

```
Python
boolean = False
```

Target Type	Code	Output	Notes
int	int(False)	0	Valid
	int(True)	1	Valid
float	float(False)	0.0	Valid
	float(True)	1.0	Valid
str	str(False)	'False'	Valid
list	list(False)	Error	Not iterable
tuple	tuple(False)	Error	Not iterable
set	set(False)	Error	Not iterable
dict	dict(False)	Error	Not iterable

9. Dictionary Conversion

To convert a list to a dictionary, it must be a list of key-value tuples.

Python

```
d = [('name', 'teja'), ('batch', '22'), ('subject', 'python')]
dict(d)
# Output: {'name': 'teja', 'batch': '22', 'subject': 'python'}
```